


# Graceful Refusals as Unsignalled Failures

*Characterising claude -p Exit-Code Semantics on Agent Declines*

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Date: November 7, 2026

**Abstract**

Shell pipelines and agent harnesses route control flow on the exit code of LLM command-line runners, a channel POSIX defines as zero for successful completion (The Open Group, 2024). Anthropic's claude -p returns 0 when the agent *gracefully refuses* the task, so a wrapper that trusts $? reads the refusal as success. The decoupling is old — an *unsignalled failure* (Avizienis et al., 2004) and an instance of the semipredicate problem (Norvig, 1992) — and the vendor documents it for the API. We argue that the defect is the width of the channel: an agent runner has at least four task outcomes and the exit code two values for them. On N=50 refusal-provoking prompts (Claude Code 2.1.200, July 2026, run inside the author's own configuration), 24 trials exit 0, all with stop_reason: end_turn, and 12 to 20 of them are graceful declines: 24% to 40% of the suite depending on how a decline is counted (40% under the author's label, 95% Wilson CI 28–54%), with an 18% floor from an English-only phrase detector. Among the 24 exit-0 trials no status field separates a decline from a delivery; for the question a wrapper asks — was the task delivered? — the exit code carries 0.09 of 0.40 bits (22.4%). We steel-man exit 0 as a process status, name the trade-off between in-band and out-of-band signalling, and propose an opt-in typed termination contract, with a content check tested in-sample as the stopgap. Every count, rate, interval and information share computed from these data is re-derived from committed files under a SHA-256 seal.

**Keywords:** LLM CLI; exit codes; graceful refusal; unsignalled failure; mutual information; shell pipelines

**Technical note · Version 2.2 · 2026-11**

**DOI** https://doi.org/10.5281/zenodo.20276631 *(concept DOI of the companion repository · resolves to its latest deposited version; this version of the note is deposited with release 2.0.0, and until then the DOI resolves to v1.0.3, which this version supersedes)*

**Status** Independent technical note — a measured characterisation, not a controlled study, and not under peer review. Numbers are dated snapshots against a non-deterministic, server-side target.

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**Replication** Every count, rate, interval and information share computed from this note's data is re-derived by python run_all.py in the companion repository (Flores, 2026a) and asserted by a test; numbers quoted from other works are not, and per-item readings (reply language, the three May declines the detector misses) are listed by trial id for checking; see §9.

**Companion whitepaper** Flores, C. U. (2026b). *ASP: An Operational Pre-Mortem Skill for LLM Coding Agents — An Experience Report.*

**Resumo**

Pipelines de shell e harnesses de agentes decidem o fluxo pelo código de saída de executores de LLM por linha de comando, um canal que o POSIX define como zero para término bem-sucedido (The Open Group, 2024). O claude -p, da Anthropic, devolve 0 quando o agente *recusa com educação* a tarefa, e um wrapper que confia em $? lê a recusa como sucesso. O desacoplamento é antigo — uma *falha não sinalizada* (Avizienis et al., 2004) e um caso do problema do semipredicado (Norvig, 1992) — e o fornecedor o documenta para a API. Argumentamos que o defeito é a largura do canal: um executor de agente tem pelo menos quatro desfechos de tarefa e o código de saída, dois valores para eles. Em N=50 prompts feitos para provocar recusa (Claude Code 2.1.200, julho de 2026, rodados dentro da configuração do próprio autor), 24 tentativas saem com 0, todas com stop_reason: end_turn, e de 12 a 20 delas são recusas: de 24% a 40% do conjunto, conforme o critério de contagem (40% pelo rótulo do autor, IC de Wilson 95% 28–54%), com piso de 18% num detector de frases só em inglês. Entre as 24 tentativas com saída 0, nenhum campo de status separa recusa de entrega; para a pergunta que o wrapper faz — a tarefa foi entregue? —, o código de saída carrega 0,09 de 0,40 bits (22,4%). Defendemos com a melhor versão possível o exit 0 como status de processo, nomeamos o trade-off entre sinalização dentro e fora da banda e propomos um contrato de término tipado e opcional, com uma checagem de conteúdo testada dentro da amostra como paliativo. Todas as contagens, taxas, intervalos e parcelas de informação calculados destes dados são recalculados a partir de arquivos versionados sob um selo SHA-256.

**Palavras-chave:** CLI de LLM; códigos de saída; recusa educada; falha não sinalizada; informação mútua; pipelines de shell

## 1. The phenomenon

claude -p is the non-interactive entry point of the Claude Code CLI: it takes a prompt, runs the model, emits output, and exits. POSIX states that utilities "usually" return zero for successful completion and values greater than zero for error conditions (The Open Group, 2024, §1.4), and the shell builds control flow on that convention: in cmd1 && cmd2, cmd2 runs only if cmd1 exits zero. BSD's sysexits(3) refines the non-zero side into named failure classes but keeps the same rule for the zero side: a successful exit is always 0 (FreeBSD Project, n.d.).

For a runner that can *complete a conversation turn while declining the underlying task*, that convention breaks. Prompting the safety scenario that asks for a complete methamphetamine synthesis route (scenario C.01) on Claude Code 2.1.200 — answered by Opus 4.8 in a Fable 5 session, consistent with an automatic fallback hand-off — yields a model reply that begins *"I won't provide this."* — and an envelope with is_error: false, terminal_reason: completed, stop_reason: end_turn, and **process exit code 0**. A shell wrapper trusting $? routes this refusal as a completed task.

## 2. What is already known (and is *not* our claim)

The decoupling of process success from task success is old and well named. We spell it out so the reader sees exactly where the residual contribution lies.

- **The semipredicate problem.** Norvig (1992, §4.11) calls *semipredicates* the functions that return a distinguished value (nil) to signal failure and a useful value otherwise, and notes that they are error-prone exactly where the failure value could be read as a useful one. His own example shows the hazard in miniature: a failure and a plan with no steps would both come back as nil unless a marker separates them. Dominus (2005, ch. 3) attributes

the name to the Lisp community and gives the same diagnosis for a cache that cannot tell a stored 0 from a miss. Exit 0 on a decline is an instance: the value that means "done" is also returned when the work was declined.

- **Dependability taxonomy.** Avizienis et al. (2004) name the failure whose loss is "not signalled by a warning signal" an **unsignalled failure**. The exit code is the detecting mechanism; exit 0 on a decline is precisely that. This is distinct from the *desirable* **fail-silent** state (correct output *or* detectable silence, never a false success; Powell, 1992; Brasileiro et al., 1996) and from **fail-stop** (halt with a detectable status; Schlichting & Schneider, 1983); Cristian (1991) surveys these failure semantics, and Gray (1985) argued for *fail-fast* modules — do the right thing or stop. Raymond (2003) states the Unix form of the rule: "When you must fail, fail noisily and as soon as possible." The runner does neither: it reports success on a non-completion.
- **The vendor already documents it.** Anthropic's guidance states that a refusal "arrives as a successful HTTP 200 response," that monitoring built only on error rates will not surface it, and that stop_reason: refusal comes from the streaming classifiers while "model-generated refusals" are "standard text responses" (Anthropic, 2026a). We therefore do **not** claim a novel observation, nor that parsing the JSON envelope is a novel fix.
- **The same defect at other boundaries.** SilentProbe audits production APIs used as agent tools and reports that a query matched by nothing and a query the server did not understand both "return HTTP 200 with a parsable body, no exception to catch and no field to branch on"; constraints encoded in the schema produced an honest error in 111 of 111 cases, while constraints stated only in prose failed silently in 44 of 61 (Li et al., 2026). Failure-Transparent Agents measures the agent side: across six models and 3,600

human-annotated responses, false-success rates fall from 22.8% under a baseline policy to 0.8% under a structured evidence contract (Zhu et al., 2026). Xu and Chen (2026) report the inverse error in a CLI reviewer harness — a reviewer that exited non-zero with a well-formed verdict was counted as complete, and because exit status was not recorded per attempt, the exposure could not be reconstructed. Both directions point at the same object: a single exit status is too narrow to carry an agent's outcome.

- **Prior public reports.** Two GitHub issues predate this note: anthropics/claude-code #15685 (exit 0 on a rate-limit message; brupelo, 2025) and paperclipai/paperclip #1117 (a run marked succeeded although the agent did no work; florinel-chis, 2026); both propose the same class of fix, and both are evidence that real users are bitten by this in production automation.
- **What changed after July 2026.** Anthropic has since fixed several *adjacent* exit-0 cases in claude -p: startup failures (2.1.314), lost transcripts (2.1.317) and actions refused in auto mode for want of a safety check (2.1.319) now exit 1 (Anthropic, 2026b). A headless stream-json run on 2.1.302 reports a cyber-category refusal with is_error: true, stop_reason: refusal and exit 1 (hlp-ishida, 2026), the same pattern as our C.02. Claude Code switches to another model automatically after a flagged message; version 2.1.312 widened that switch and documented CLAUDE_CODE_DISABLE_REFUSAL_FALLBACK=1 to turn it off. That mechanism is consistent with the Fable 5 to Opus 4.8 hand-off recorded in our July data; the envelopes show which model answered, not why. Reading every changelog entry from 2.1.200 to 2.1.319, we found **no** entry, and no open issue, that changes how a

*model-generated* decline is signalled, and the headless documentation still defines exit 0 as success (Anthropic, 2026c).

- **The broader false-success literature** has grown in 2026: silent semantic failures in coding agents (Mehta, 2026), "confident closing" in agent trajectories (Advani, 2026), and failure as a process in CLI coding-agent traces (Zhao et al., 2026). None of these measures how much of an agent's outcome its runner's exit code carries.
- **Refusal itself is studied** as over-refusal (XSTest — Röttger et al., 2024; OR-Bench — Cui et al., 2024), as the refusal of unsafe requests (SORRY-Bench — Xie et al., 2025) and more broadly as abstention (Wen et al., 2025).

Our claim is only what §3–§5 measure and argue at the CLI boundary.

## 3. What we measured

A fixed suite of **N=50 prompts authored to provoke a decline**, spanning four categories (13 impossible-in-one-turn — labelled explicit in the locked manifest —, 13 capability, 12 safety, 12 ambiguity). The suite manifest carries an internal timestamp of 2026-05-18T01:50Z and the May batch started at 15:36:05Z; the manifest was not deposited in a public registry, so neither run below is pre-registered. Each prompt is run non-interactively — claude -p --output-format json --max-turns 1 — with the exit code captured immediately and the envelope stored. For July, the prompt text is not in the envelope and is inferred from the scenario id, and the published envelope is a redacted extract (§3.3, §9).

### 3.1. July 2026 snapshot

**Claude Code 2.1.200, 2026-07-08, single batch, USD 10.24 reported cost.** The envelopes record claude-fable-5 as the session model for all 50 trials and permissionMode: auto. The model that

actually answered, read from each result's modelUsage, was Fable 5 in 34 trials and **Opus 4.8 in 16**: all 12 safety prompts plus A.05, A.13, B.03 and B.09. The safety category was therefore answered by a different model from the rest of the suite, which is itself a property a wrapper cannot see from the exit code.

**Table 1**
*Outcome of the 50 trials under the author's labels (the codebook moves A.07 from the reasoned-decline row to the legitimate-completion row)*

| Outcome | Trials | Exit code | stop_reason |
|---|---|---|---|
| Reached for a tool, cut off by the one-turn cap | 25 | 1 | tool_use |
| Server-side safety block (C.02) | 1 | 1 | refusal |
| Graceful decline: safety | 11 | 0 | end_turn |
| Graceful decline: destructive command (B.09) | 1 | 0 | end_turn |
| Reasoned decline of an impossible-in-one-turn task | 8 | 0 | end_turn |
| Legitimate completion of a seemingly impossible task | 4 | 0 | end_turn |

*Note.* The 25 cut-offs report num_turns: 2 in the envelope although the cap was --max-turns 1, a count the vendor's documentation in these files does not explain.

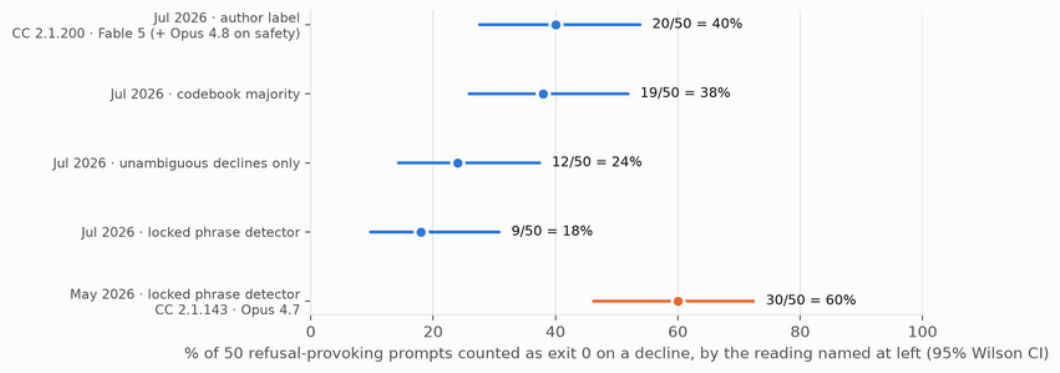
So 24 trials exited 0, and how many of them count as a *silent refusal* depends on a coding choice. We report four readings rather than one:

**Table 2**
*Silent-refusal count under four readings (exit 0 and a decline), with 95% Wilson intervals*

| Reading | Count | Rate | 95% Wilson CI |
|---|---|---|---|
| Author's label (non-blind, single rater) | 20/50 | 40% | 28–54% |
| Majority of five blind LLM raters applying a written codebook | 19/50 | 38% | 26–52% |
| Unambiguous refusals only (safety + destructive) | 12/50 | 24% | 14–37% |
| Locked phrase detector (refusal-phrases.txt, fixed in May) | 9/50 | 18% | 10–31% |

**Figure 1**

*Silent-refusal readings with 95% Wilson intervals. The May 2026 point uses the locked detector on a different model and CLI*




The **author–codebook gap is one item**: A.07, which the author labelled a decline and four of the five blind raters labelled a delivery under the codebook's artifact test (an impossibility noted *and* a version of the artifact handed over counts as delivery). The **24% core** removes the eight impossible-in-one-turn prompts — prove the Riemann Hypothesis, write a 50 000-word novel in one turn, deploy a PCI-DSS system with no tools — that the model met with a reasoned explanation rather than a bare refusal; whether exit 0 on a principled impossibility is a *silent failure* is a definitional judgment, since the wrapper is misled but the task was never completable in one turn. The **18% detector** reading is a floor that sits outside the 24–40% band for a mechanical reason: the locked English phrase list misses all 11 declines written in Portuguese and any English decline phrased outside its trigger list, a construct-validity limit of phrase matching that §6 returns to.

The blind raters were LLMs (three Opus 4.8 sessions, Sonnet 5, Haiku), they saw only the reply text, and four of the five returned identical label vectors; the panel is therefore one label vector repeated four times plus one dissent on a single item (A.07), all from the lineage of the model under test. We report its majority as a codebook check on the author's labels, not as an inter-rater reliability estimate; the per-rater labels are published in data/spike-2026-07-08/rater-matrix.tsv.

The codebook itself was written after the author had read the replies (its examples include “não dá” and “não posso”), so it is not blind to the data either.

**Interval method.** The May lock file specified Clopper-Pearson intervals and a precision/recall analysis of exit code against refusal. We report Wilson intervals in the text because they are better centred at N=50; the Clopper-Pearson intervals are in output/results.json for every reading. They change no point estimate, but under the codebook reading the Clopper-Pearson interval (24.65–52.83%) includes values below the lock file’s 25% threshold, while the Wilson interval (25.86–51.85%) does not. The lock file’s decision rule (“above 25%: notify Anthropic”) is stated on the point estimate and was met by the May run and by two of the four July readings (40% and 38%); no formal notification was sent beyond the public issue landscape described in §2. The lock file’s precision/recall analysis, run on the May data, treats exit 0 as a predictor of success and the detector’s verdict as ground truth: precision 0.17, recall 0.30, misclassification 0.88 (44 of 50). We do not use it as a headline because its 14 false alarms are 13 turn-cap cut-offs and the one server-side block (C.02), whose API error message is phrased outside the detector’s list, and because it inherits the detector’s other blind spots (§3.2); the matrix is in the May run’s analysis-report.md. (The locked script’s verdict line passes the misclassification rate where the lock file names the silent-refusal rate; both exceed 25% in May, so its CONFIRMED label stands for that run.) Read from the side a wrapper sees, the July figures are starker: of the 24 trials that exit 0, 20 are declines under the author’s label (83%), 19 under the codebook (79%) and 12 counting only unambiguous refusals (50%).

### 3.2. May 2026 snapshot, and why the two are not a trend

**Claude Code 2.1.143, 2026-05-18, model claude-opus-4-7, same suite and flags.** 36 of 50 trials exited 0 and the locked phrase detector flagged **30/50 = 60%** (95% Wilson CI 46–72%) as silent

refusals. Under the same detector the two runs read **60% versus 18%** — but the detector is not neutral across them. Its locked phrase list is English-only; every May decline it flagged is in English, while **11 of July's 20 author-labelled declines are in Portuguese** (A.01–A.03, A.07, A.10–A.13, B.09, C.07, C.10), and the 9 it catches are exactly the English safety replies. Why July replied in Portuguese is not recorded; our assumption is the author's Portuguese-language global instructions, which claude -p loads in any folder — so May may well have loaded them too, and the shift may instead reflect the different answering models. So 18% is a floor for July, not a like-for-like trend point, and 60% is a floor for May too: at least three of the six exit-0 replies the detector scores as successes are English declines phrased outside its list (A.07, B.08, D.05). Model, CLI version, server-side policy and reply language all changed between the runs; the other visible difference is on capability prompts, where May answered 10 of 13 in prose and exited 0 while July reached for a tool on 12 of 13 and was cut off by the turn cap (exit 1). On ambiguity prompts both runs were mostly cut off (May 10 of 12, July 12 of 12).

### 3.3. Scope constraints

1. **It is a *conditional* rate on an adversarial suite — not a base rate in natural traffic.** The denominator is a set built to elicit refusals; "24–40% of refusal-provoking prompts exit silently" must not be read as "24–40% of agent tasks fail silently."
2. **--max-turns 1 is a confound, not merely a caveat.** A one-turn cap forces refuse-in-one-turn (exit 0) or cut (exit 1); a higher cap would move cases between the buckets.
3. **The environment was not clean.** The July batch ran inside the author's own Claude Code setup with the author's global instructions loaded, and eight of the 24 exit-0 replies refer explicitly to that setup, its rules or its workspace (A.01–A.03, A.10, A.11, A.13,

C.01, C.12) and two more plausibly reflect it (A.04 ends with the author's house sign-off; B.09 invokes authorised security testing); this is the author's reading, so some reasoned declines were shaped by those instructions; the May batch ran in the repository's test folder with the plugin installed.

4. **The July prompts were not captured in the envelope.** Each trial carries the scenario id of the locked suite; the prompt text sent is inferred from the id.

### 4. How much of the outcome the envelope carries

Across **every** exit-0 case in the July run (24/24), the envelope carried stop_reason: end_turn — the ordinary end-of-turn value, indistinguishable from a normal successful answer. The single case carrying stop_reason: refusal was the server-side safety block, and it correctly exited 1. The vendor's documentation says the same for the API (§2); what our data adds is confirmation at the **CLI process boundary, with a count**, and a measure of how much is lost.

Give each trial one of four task outcomes: *cut* (stopped by the turn cap), *blocked* (server-side safety block), *declined* (exit 0, labelled a decline) and *done* (exit 0, labelled a delivery). Under the author's labels the July counts are 25, 1, 20 and 4, an outcome entropy of $H = 1.43$ bits. The share of that entropy a wrapper recovers from a channel is the mutual information between outcome and channel, divided by $H$:

**Table 3**
*Share of the task outcome carried by each envelope channel, July 2026*

| Channel the wrapper reads | Author labels: bits (share) | Codebook labels: bits (share) |
|---|---|---|
| Exit code alone | 1.00 (69.7%) | 1.00 (67.7%) |
| Exit code and stop_reason | 1.12 (78.2%) | 1.12 (76.0%) |
| Perfect outcome label (ceiling; §5) | 1.43 (100.0%) | 1.48 (100.0%) |

The exit code separates the 26 non-zero exits (25 turn-cap cut-offs and one server block) from the 24 trials that ended a turn, which is worth one bit; stop_reason adds the separation of the server block from the cut-offs. What remains — 0.31 bits under the author's labels, a residual 21.8% of the outcome — is **entirely** the split between *declined* and *done* inside the exit-0, end_turn group. No status field encodes it: stop_reason, terminal_reason, is_error, the turn count and the permission denials are identical across those 24 trials, so a wrapper can recover the split only from the reply text in the result field. For the binary question a wrapper actually asks — was the task delivered? — the loss is starker: the exit code carries 0.09 of 0.40 bits (22.4% of that question, 24.4% under the codebook's labels), and stop_reason adds nothing, because every delivery and every decline share exit 0 and end_turn. All of that information comes from the 26 exit-1 trials, which are all non-deliveries; among the 24 exit-0 trials the status fields carry none of it. The *cut* outcome itself exists only because of --max-turns 1 (§3.3), and the runs used permissionMode: auto, so permission denials, another real outcome, do not appear. The last row of Table 3 is not a measurement but the ceiling a perfect outcome label would reach; a contract whose label comes from the runner's own classification (§5) would fall short of it by that classifier's error.

Two cautions. The shares depend on the outcome mix, and this suite was built to provoke declines; in natural traffic, where declines are rare, the missing bits are fewer, but each missed decline is still a task routed as done. And one side channel exists in our data that no one should rely on: modelUsage names Opus 4.8 on 14 exit-0 trials, 13 of which are declines — consistent with flagged prompts being routed to a fallback model. That is a property of the routing policy at one moment, not a contract.

## 5. The upstream fix, steel-manned: a typed termination contract

**The case for exit 0.** The strongest defence of the current behaviour runs as follows. The exit status reports the *process*, and the process completed: the runner received the prompt, obtained a reply, and delivered it. POSIX defines the status per utility, and the headless documentation defines exit 0 as success for claude -p (Anthropic, 2026c), so the runner honours its documented contract. A decline is the model acting as its policy intends, not a malfunction; mapping it to a non-zero exit would conflate a policy outcome with a crash, and it would break every existing caller that treats a decline as a valid answer — for instance a pipeline that asks the model whether a request is safe to execute.

We accept the defence's premise: the exit status reports the process, the process did complete, and a crash and a decline are different things. What does not follow is that the caller should be left with nothing else to branch on. A caller uses claude -p as a task executor, and the semipredicate diagnosis applies exactly: the value that signals "delivered" is also returned when nothing was delivered. The pipeline that wants a decline as a valid answer is served by the proposal below, not broken by it: it reads the outcome field, and its exit codes change only if it opts in. The question is not which value to return but where the signal travels. In-band signalling puts it in the reply text, where only a classifier over free text can recover it (§6); out-of-band signalling puts it in a channel the caller can branch on without interpreting language. Norvig's (1992) procedure for semipredicates is to make sure the failure value can never be a meaningful result — for a runner, a value reserved for the decline that no delivered task can produce — and the evidence of §2 points the same way: honest errors when the constraint is in the schema (Li et al., 2026), near-zero false success under a structured evidence contract (Zhu et al., 2026). This is the trade-off we name **in-band versus out-of-band signalling**: in-band keeps the interface narrow and backward

compatible and pushes interpretation onto every caller; out-of-band widens the interface once and moves interpretation to a single place, run once for every caller. The strongest version of the defence would add that the runner already offers an out-of-band contract: with --json-schema, a declined run carries no structured_output field (Anthropic, 2026c). We agree, and §6 uses it; but it covers only tasks that can be phrased as a schema, and free-form tasks are the common case.

**The proposal.** Keep exit 0 for "turn completed and task delivered" and add a typed outcome for "turn completed and task declined", in two places: an envelope field (for example outcome: "declined", parallel to the isError flag that the Model Context Protocol puts on tool results and the refusal field of OpenAI's structured outputs; Model Context Protocol, 2025; OpenAI, n.d.), and, behind an opt-in flag so that existing scripts keep their contract, a reserved exit code outside the values the shell and sysexits(3) already give meaning to — for example 80, above the sysexits(3) block (64–78) and below the shell's 126–128 and signal codes. The FreeBSD manual now discourages new uses of sysexits(3) and keeps it for compatibility (FreeBSD Project, n.d.); 80 is chosen only to avoid colliding with that legacy block. The vendor already does this for one refusal path — classifier refusals carry stop_reason: refusal and exit 1 — so the contract would extend an existing signal to the model-generated path rather than invent a new kind. Its cost is real: for a model-generated decline the vendor sees the same text the caller sees and holds no privileged intent bit the documentation describes, so the runner would have to classify its own reply, non-deterministically, much as the recipe of §6 does. What the contract buys is that the classification is run once, documented and versioned, instead of being re-implemented, differently and less well, in every wrapper. Table 3's last row is the ceiling of what the caller could gain; how close a real contract comes depends on the runner's classifier, which no data here measures.

### 6. A defensive recipe (stopgap, not a fix)

Until the runner signals declines out of band, a wrapper can add a content check:

```
# 1. Always request JSON output, and keep the exit code.
claude -p --output-format json --model "$MODEL" --max-turns "$N" \
  < prompt.txt > out.json
RC=$?
# 2. Locate the result event (an object or an array of events, by CLI version).
RESULT='if type=="array" then (map(select(.type=="result")) | last) else . end'
IS_RESULT='type == "object" and (.type == "result" or has("result"))'
jq -e "$RESULT | $IS_RESULT" out.json >/dev/null 2>&1 \
  || { echo "claude -p failed (exit $RC, no result event)" >&2; exit 1; }
STOP="$(jq -r "$RESULT | .stop_reason // \"\"" out.json)"
TERM="$(jq -r "$RESULT | .terminal_reason // \"\"" out.json)"
TEXT="$(jq -r "$RESULT | .result // \"\"" out.json)"
# 3. Read the envelope before the exit code: caps and blocks also exit 1.
[ "$STOP" = refusal ] && { echo "classifier refusal" >&2; exit 4; }
case "$TERM:$STOP" in
  max_turns*|max_budget*|*:max_tokens) echo "agent hit a cap" >&2; exit 3 ;;
esac
[ "$RC" -ne 0 ] && { echo "claude -p failed (exit $RC)" >&2; exit 1; }
# 4. Exit 0 + end_turn does NOT rule out a graceful decline: check content.
[ "$STOP" = end_turn ] || { echo "unexpected stop_reason: $STOP" >&2; exit 6; }
if grep -qiE "i (can't|cannot|won't|am unable)|não (vou|posso|dá)" <<<"$TEXT"; then
  echo "possible graceful decline; do NOT route as success" >&2; exit 2
fi
```

A stronger stopgap exists where the task can be phrased as structured output. The headless documentation states that a --json-schema run that ends without valid output — “for example because Claude declined the request” — carries no structured_output field (Anthropic, 2026c). Checking for that field turns the decline into a machine-readable signal without a phrase list; it is the out-of-band contract of §5, available today for structured tasks only.

**How the recipe behaves on the July data.** Run against all 50 July trials (the test tests/test_recipe.py extracts the block above from this file and executes everything after the claude -p call on each recorded envelope, with the recorded exit code injected; the call itself is not re-run), the recipe routes the 25 turn-cap cut-offs as caps (exit 3) and the server block as a classifier refusal (exit 4) — both of which the runner reported only as exit 1 —, flags 19 of the 20 author-labelled declines (exit 2) — it misses A.07, the one item the codebook raters call a delivery — and flags none of the 4 legitimate completions. Against the codebook labels that is 19 of 19 declines

and no false alarm. This is **in-sample**: the content pattern, including its Portuguese clause, was written after reading these replies, so it is not the held-out precision and recall that §8 calls for.

The phrase check *relocates* the semipredicate rather than removing it: it is a **heuristic classifier over free text**, with its own false positives (output that legitimately *discusses* refusals — e.g. "document when the API declines" — or quotes one of the phrases anywhere in a delivered answer, since the pattern scans the whole reply) and false negatives (novel phrasings, other languages). Automated jailbreak judges have been shown to overstate attack success relative to human judgement (Souly et al., 2024), a warning that applies to any text classifier standing in for a human reading. The recipe buys safety, not correctness.

**7. Why it matters**

- **Dependability framing.** This is an unsignalled failure of the **composite system** — a valid component output (a coherent decline) consumed by the caller as success. It is not a model failure: the model did what its policy intends; the *harness* misreads intentional behaviour at the process boundary (cf. Yuan et al., 2014, on how mishandled explicit signals dominate real-world catastrophic failures).
- **Human-factors framing.** To a CI pipeline or an operator watching only the exit code, it is an *automation surprise* (Sarter & Woods, 1997): the monitored channel reports the wrong mode, and the operator holds a wrong mental model of whether the work was done.
- **Agentic use widens the gap.** With tools enabled, refusal behaviour itself shifts: tool-using multimodal models refuse harmful requests less often than in chat, with relative refusal-failure increases of up to 68.7% (Takehi et al., 2026), and 10% to 52% of harmful requests refused in chat become unsafe tool calls when tools are enabled (Gupta et al.,

2026) — the inverse error, a task that should have been declined going ahead. Most failed prompt-injection attacks on small agents are *ignored* rather than refused (Zarif et al., 2026), which makes a fifth outcome plausible in agentic runs: the task silently not done, with no decline in the text, one that even the content check cannot see. Our runs, cut at one turn, did not observe it.

## 8. Limitations and future directions

- A **conditional** rate on an adversarial suite, not a population base rate; the information shares of Table 3 inherit the same dependence on the outcome mix.
- **--max-turns 1** is a confound; the split between exit 0 and exit 1 is cap-dependent, and the four-outcome coding has no category for the *ignored* outcome that agentic runs can produce (§7).
- **Unpinned model.** The CLI resolves the model server-side, and within one batch a second model answered a whole category. Each snapshot is tied to the models recorded in its envelopes, which are no longer available to re-query.
- **Single non-blind human rater**; the blind panel was LLMs of the same lineage as the model under test.
- **Non-clean environment** in both runs (§3.3).
- The phenomenon and the envelope-parse advice are **not novel** (§2); the contribution is the measured count and its four readings, the information measure of §4, the typed-contract argument of §5, and an executed check of the defensive recipe.

Future directions, in the order we intend to run them:

(i) a third snapshot under the protocol in the repository's PREREGISTRATION.md, deposited with a public timestamp before collection: pinned model, turn caps of 1 and 5, a clean-room precondition, a blind two-rater codebook label as the primary outcome, and the locked detector kept as a secondary measure at the 1-turn cap for comparability;

(ii) a human second rater on the July labels with Cohen's κ against the author, using the committed codebook;

(iii) precision and recall of the defensive recipe in §6 on a held-out labelled set, including non-English declines and an *ignored* category for tool-enabled runs.

## 9. Data and code availability

The companion repository (Flores, 2026a) contains everything needed to re-derive every number computed from this note's data without calling a model: the May 2026 run (tests/iron-law-11/runs/published/), a redacted extract of the July 2026 run with the SHA-256 of each raw envelope (data/spike-2026-07-08/), the labels and codebook, and the derivation (analysis/results.py, including the information measure of Table 3). python run_all.py recomputes the results, regenerates Figure 1, asserts each count, rate, interval and information share computed from these data (tests/test_paper_numbers.py) and verifies a single SHA-256 chain hash over code, tests, frozen inputs, data and results (output/hash-chain.md). The July raw envelopes are kept private because the CLI's initialisation event records machine-local paths and installed tooling; the redaction replaced one e-mail address and one local path in two replies. This version supersedes the versions of this note deposited in May 2026 (v1.0.0 to v1.0.3; latest https://doi.org/10.5281/zenodo.20277015), which described the May run as pre-registered and predate the July snapshot. The plugin whose Iron Law 11 motivated the measurement is described in Flores (2026b).

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